

Mathematics: Calculus

Middletown High School | Grade 12, Fall Semester 2026

Teacher: Ms. A. Whitfield
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Course Description

A first course in calculus for seniors, equivalent to a first semester college course. Students work with limits, derivatives and integrals, and apply them to real problems using graphs, tables, algebra and clear written reasoning.

Course Units

Weeks 1 to 2	Ch1: Algebra Foundations Linear equations, inequalities and systems. Solving for variables, graphing lines in the form $y = mx + b$, substitution and elimination.
Weeks 3 to 5	Ch2: Quadratic Functions Standard form $ax^2 + bx + c = 0$ and vertex form $y = a(x - h)^2 + k$. The discriminant $b^2 - 4ac$ decides the number of real roots. Factoring, completing the square and the quadratic formula.
Weeks 6 to 8	Ch3: Trigonometry SOH CAH TOA, the unit circle, radians and degrees, sine, cosine and tangent with their inverses, and the identity $\sin^2(\theta) + \cos^2(\theta) = 1$.
Weeks 9 to 13	Ch4: Introduction to Calculus Limits and continuity, the derivative as a rate of change, power, product and quotient rules, and an introduction to the definite integral.
Weeks 14 to 16	Ch5: Statistics and Probability Measures of center and spread, normal distributions, conditional probability and interpreting data sets.

Key Dates This Semester

Sep 25	Ch2 Quiz, quadratic functions
Oct 20	Mid-Term Exam, covers Ch1 to Ch3
Nov 13	Ch4 Assignment due, derivatives in context
Dec 15	Final Exam, all chapters

School wide dates

Sep 8	First day of classes
Oct 19 to Oct 23	Mid-term examination week
Nov 6	Parent teacher conferences
Nov 25 to Nov 27	Thanksgiving break, no classes
Dec 14 to Dec 18	Final examination week

Materials

Graphing calculator (TI-84 or equivalent), three ring binder, graph paper, pencils.

Grading Policy

15 percent	Homework
15 percent	Classwork and participation
20 percent	Quizzes
25 percent	Unit tests
25 percent	Final exam

A = 90 to 100 percent, B = 80 to 89, C = 70 to 79, D = 60 to 69, F = 59 and below.

Late and Make-Up Work

Work handed in late without an excused absence receives a maximum of 50 percent credit. With an excused absence,

work is due within one week of returning. It is the student's responsibility to ask what was missed. Make-up tests are held on Thursdays after school and must be booked at least one day ahead.

Academic Honesty

All work submitted must be your own. Copying another student's work, or submitting text or code generated by someone or something else as your own, is treated as academic dishonesty and results in a zero for that assignment and a call home.

Extra Help

Extra help is available Tuesdays and Thursdays, 3:15 to 4:00 pm, in the room listed above.